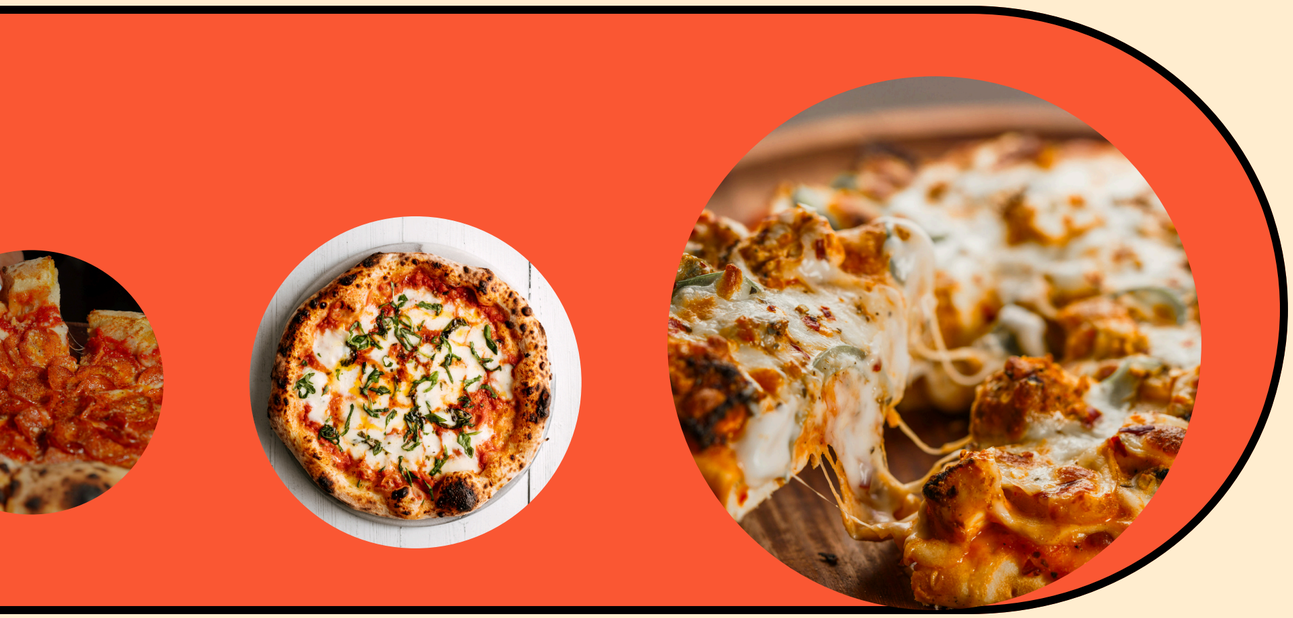
PIZZA SALES PROJECT

SqL







HELLO

"Hi everyone,
I'm Harsh.

For this project, I explored a pizza sales dataset using SQL to find out what customers love to order and when sales peak.

By writing SQL queries, I was able to turn raw data into simple answers—like which pizza sells the most and what times are the busiest. Let's take a quick look at what the data revealed!"



```sql
-- NOW 1STR QUESTION
-- Retrieve the total number of orders placed.

SELECT
    COUNT(order_id) AS TOTAL_ORDERS
FROM
    ORDERS;

-- NOW QUESTION 2
 -- Calculate the total revenue generated from pizza sales.
 -- FOR REVENUE TOATL PIZZAS * PRICES

SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_sales
FROM
    order_details
        JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id;
```

Result Grid | Filter R

	TOTAL_ORDERS
▶	2833

Result Grid | Filter Rows:

	total_sales
▶	817860.05

```sql
43
44          -- NOW QUESTION 3
45          -- Identify the highest-priced pizza.
46
47          SELECT
48          pizza_types.name, pizzas.price
49      FROM
50          pizza_types
51              JOIN
52          pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
53      ORDER BY pizzas.price DESC;
54
55      -- NOW QUESTION 4
56      -- Identify the most common pizza size ordered.
57
58      SELECT quantity , count(order_details_id)
59      FROM order_details GROUP BY quantity ;
60
61      SELECT
62          pizzas.size, COUNT(order_details.order_details_id)
63      FROM
64          pizzas
65              JOIN
66          order_details ON pizzas.pizza_id = order_details.pizza_id
67      GROUP BY pizzas.size;
```

Filter Rows:

COUNT(order_details.order_details_id)
15385
18526
14137
544
28

Result Grid | Filter Rows:

name	price
The Greek Pizza	35.95
The Greek Pizza	25.5
The Brie Carre Pizza	23.65
The Italian Vegetables Pizza	21
The Spinach Supreme Pizza	20.75
The Barbecue Chicken Pizza	20.75
The California Chicken Pizza	20.75
The Spicy Italian Pizza	20.75





```sql
-- NOW QUESTION 5
-- List the top 5 most ordered pizza types along with their quantities.

SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC;
```

Result Grid | Filter Rows:

name	quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371
The California Chicken Pizza	2370
The Sicilian Pizza	1938
The Spicy Italian Pizza	1924

Result 52

```sql
-- NOW Intermediate LEVEL
-- QUESTION 1 Join the necessary tables to find the total quantity of each pizza category ordered.

SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;

-- QUESTION 2
-- Determine the distribution of orders by hour of the day.

SELECT
    HOUR(order_time), COUNT(order_id)
FROM
    ORDERS
GROUP BY HOUR(order_time);
```

Result Grid | Filter Rows:

category	quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050



Result Grid | Filter Rows:

HOUR(order_time)	COUNT(order_id)
11	169
12	318
13	297
14	246
15	192
16	243
17	322
18	314

```sql
-- QUESTION 3: Join relevant tables to find the category-wise distribution of pizzas.

SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;

-- QUESTION 4 : Group the orders by date and calculate the average number of pizzas ordered per day.

SELECT
    round(AVG(quantity),0)
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

Result Grid | Filter Rows

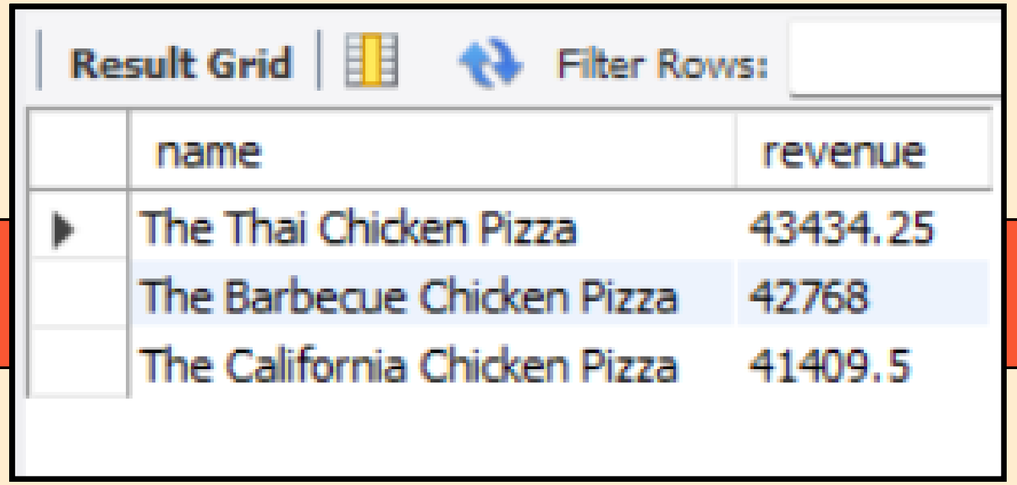
category	COUNT(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

Result Grid | Filter Rows:

round(AVG(quantity),0)
136

```sql
-- QUESTION 5
-- Determine the top 3 most ordered pizza types based on revenue.

SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```



name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5



```sql
-- Advanced level Questions
-- QUESTION 1  Calculate the percentage contribution of each pizza type to total revenue

SELECT
    pizza_types.category,
    SUM(order_details.quantity * pizzas.price) /  ( SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_sales
FROM order_details JOIN
    pizzas ON pizzas.pizza_id = order_details.pizza_id)*100 AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
        JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

	category	revenue
▶	Classic	26.9059602556967
	Supreme	25.4563126009862
	Chicken	23.955137556847287
	Veggie	23.682590927384577

```sql
  --  QUESTION 2
-- Analyze the cumulative revenue generated over time.

select order_date,
sum(revenue) over ( order by order_date) as cum_revenue
from
 (select orders.order_date ,
 round(sum(order_details.quantity*pizzas.price),2) AS revenue
FROM order_details join pizzas
on order_details.pizza_id = pizzas.pizza_id
join orders
on orders.order_id = order_details.order_id
group by orders.order_date) as sales_by_day ;
```

Result Grid | Filter Rows:

order_date	cum_revenue
2015-01-01	2713.85
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05

```sql
-- question 3
-- Determine the top 3 most ordered pizza types based on revenue for each pizza category.

select name,revenue
from
(select category,name,revenue,rank() over (partition by category order by revenue desc) as rnk
from
(SELECT
    pizza_types.category,
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM pizza_types JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
    JOIN
order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category , pizza_types.name) as A ) as B
where rnk<= 3;
```

Result Grid | Filter Rows:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25
The Pepperoni Pizza	30161.75
The Spicy Italian Pizza	34831.25
The Italian Supreme Pizza	33476.75

Result 60


THANK YOU